



Google Assistant's new feature

Google Assistant has replaced "What's on my screen?" button with a Lens-branded shortcut. <https://digit.in/sep22-21>



VI customers' cell data leaked

A cyber-security research firm claimed that call data records of 20 million customers of VI were leaked. <https://digit.in/sep22-22>

Credit: CounterComplex



An image depicting the resemblance of the different ancient number systems

ated with activity." While making the observation, Viznut made it clear that whether or not Leibniz had some inspiration from the ancient world. There were definitely a lot of similarities in the way that humans worked towards progress and the binary system was no different.

Before we dive deep into the hows and whats of the numbers though, let us understand the underlying principles of the digits we know as 0s and 1s. We have covered the origins of the binary numbers back in November 2020 and you can check it out if you want at: <https://digit.in/BinaryNosOrigins>

Basically, instead of the regular decimal system where we have ten symbols to represent the various numbers, in the binary system, we have only 2 of those said symbols in the form of the 0 and the 1, and believe it or not, we can represent any and every number using only these two puny yet powerful symbols! Anyway, we're not here to discuss

the specifics of the language that logic gates use rather we are here to focus on the small devices that help us perform almost every task that requires a technical device.

THE FIRST FEW USES OF LOGIC BASED SYSTEMS

The first logic-based machine came around in the form of Charles Babbage's Difference Engine. It used mechanical gears as processors for the logic that it needed to perform calculations.

It can also be credited to be the first-ever device to use an actual logic gate. Though Babbage later on also designed the Analytical Engine which would also use similar mechanisms. But that never got built until the 1980s as Babbage failed to secure funds to make it himself.

However, the first more commonly used component which functioned as a logic gate was the electromagnetic relay made by Joseph Henry back in 1835 which surprisingly went unno-

ticed until it proved its usefulness by being used in the telegraph and also in the later developed Rotary dial that used even more complex relays with 10 positions. These tubes functioned as basic on-off switches and helped in the processing of data that would have to be transversed.

These laid down the foundations of the kind of technology that would be used in the future and greatly helped in shaping the same.

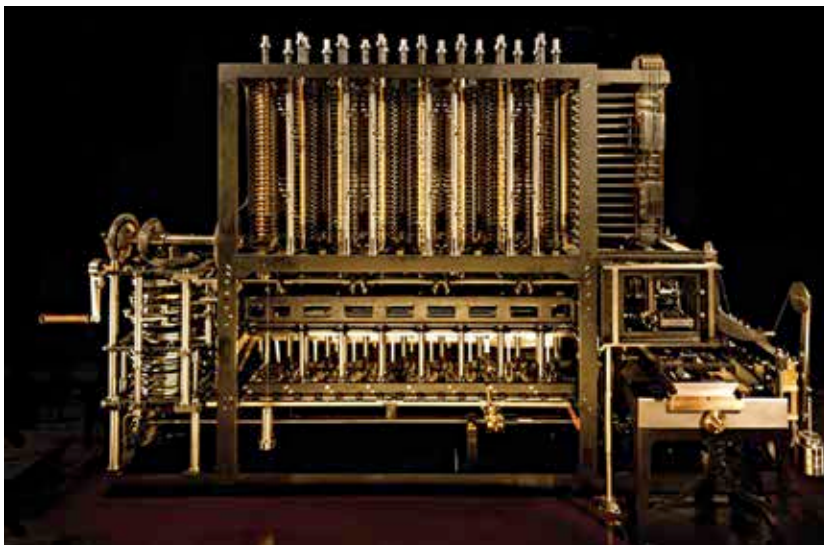
Once the electromagnetic relay found its pace however, along came one of the greatest inventors of all



Nikola Tesla

time, Nikola Tesla. Tesla, in 1898, introduced the "teleautomation", a patented technology that would allow for the usage of technology over long distances essentially creating the first ever remote controlled device. He demonstrated the technology with the usage of a remote controlled boat that used a specific method to decode Hetzian waves. It used to decode the signals it would get and perform based on the provided inputs which made it function essentially as what we know today to be an AND Gate.

After the relays, it was time for the vacuum tube to shine. These tubes had been in development over a long period of time but never for the field of computing. Not yet at least. Scientists like Edison and Tesla experimented with such tubes as they were treated as novelties except for the case of the light bulb which was slowly turning into a household object.



Charles Babbage's Difference Engine